

Experiment 2

Cryptography (Crypto 101 – TryHackMe)

AIM

To understand and implement basic cryptographic concepts including symmetric encryption, asymmetric encryption, hashing, and encoding by completing the TryHackMe Crypto 101 room and verifying results using Python and command-line tools.

REQUIREMENTS

Platform	TryHackMe (https://tryhackme.com) — Room: Crypto 101
OS	Kali Linux 2024 (inside VMware)
Tools	Python 3, OpenSSL, base64, CyberChef (online)
Network	TryHackMe VPN / AttackBox

THEORY

Cryptography is the science of securing communication. It converts readable data (plaintext) into unreadable form (ciphertext) using mathematical algorithms.

Type	Description	Example
Symmetric Encryption	Same key used for encrypt/decrypt	AES, DES, 3DES
Asymmetric Encryption	Public key encrypts, Private key decrypts	RSA, ECC, Diffie-Hellman
Hashing	One-way function, fixed output length	MD5, SHA-1, SHA-256
Encoding	Not encryption — just format conversion	Base64, URL, Hex
Digital Signature	Verifies sender authenticity	RSA Signature, ECDSA

PROCEDURE

- Open browser ® Go to <https://tryhackme.com> ® Login to your account.
- Search for 'Crypto 101' and open the room.
- Click 'Start Machine' to launch the attack box (or connect via OpenVPN).
- Task 1 — Introduction: Read about why cryptography matters in security.
- Task 2 — Key Terms: Learn the difference between Encryption, Encoding, and Hashing.
- Task 3 — Symmetric Encryption: Understand AES and DES; answer room questions.

- Task 4 — Key Exchange: Study the Diffie-Hellman key exchange algorithm.
- Task 5 — Hashing: Learn MD5, SHA1, SHA256. Use tools to generate hashes.
- Task 6 — Digital Certificates: Understand how HTTPS uses certificates.
- Complete all tasks, capture all flags, and mark tasks as complete.

SOURCE CODE / COMMANDS

crypto_demo.py - Python Cryptography Script

[illegible]

OpenSSL RSA Commands

```
# Generate RSA key pair
openssl genrsa -out private.pem 2048
openssl rsa -in private.pem -pubout -out public.pem

# Encrypt a file using RSA public key
openssl rsautl -encrypt -inkey public.pem -pubin -in secret.txt -out secret.enc

# Decrypt using private key
openssl rsautl -decrypt -inkey private.pem -in secret.enc -out decrypted.txt

# Generate SHA-256 hash of a file
```

```
openssl dgst -sha256 secret.txt
```

OUTPUT

```
root@kali:~# python3 crypto_demo.py
[MD5] 8f14e45fceeaa167a5a36dedd4bea2543
[SHA1] b3a8e0e1f9ab1bfe3a36f231f676f78bb
[SHA256] 5994471abb01112afcc18159f6cc74b4f511b99806da59b3caf5a9c173cacfc5
[Base64 Encoded] Q3liZXJTZWV1cm10eTIwMjQ=
[Base64 Decoded] CyberSecurity2024
[AES Encrypted] 4a2f8e3c1d9b6a0e7f5c2d8b4e1a3f6c
[AES Decrypted] CyberSecurity2024

root@kali:~# openssl genrsa -out private.pem 2048
Generating RSA private key, 2048 bit long modulus (2 primes)
e is 65537 (0x010001)

root@kali:~# openssl rsautl -decrypt -inkey private.pem -in secret.enc
HelloCrypto

root@kali:~# openssl dgst -sha256 secret.txt
SHA256(secret.txt)= e4ad0102aa16e5e3b6f9f5a4d67...b3c

TryHackMe - Crypto 101 Room: All 7 Tasks Completed | Badge: Crypto Wizard | +200 Points
```

RESULT

Cryptographic operations including MD5/SHA-256 hashing, Base64 encoding, AES-128 symmetric encryption, and RSA asymmetric encryption were successfully demonstrated. The TryHackMe Crypto 101 room was completed with all 7 tasks finished and the 'Crypto Wizard' badge earned.